

Cost-Sensitive Customer Churn Prediction: A Business Utility-Driven Machine Learning Approach

Mobin Abedian
Student ID: 403222152
mobinabedian6@gmail.com

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Abstract

This report presents a complete end-to-end cost-sensitive churn prediction pipeline for the Telco Customer Churn dataset. We engineer 10 custom behavioral features, evaluate 9 core classification models under SMOTE-balanced training with systematic 5-fold cross-validated hyperparameter optimization, and extend evaluation to LightGBM, CatBoost, and stacking ensemble for bonus credit. Moving beyond traditional accuracy-oriented metrics, we optimize decision thresholds against an asymmetric business utility function $U = 100 \cdot TP - 20 \cdot FP$, sweeping $\tau \in [0, 1]$ in steps of 0.01. AdaBoost achieves the highest net profit (20,040 at $\tau^* = 0.44$), while Logistic Regression ($\tau^* = 0.26$, 19,920) is selected as the final model for its superior interpretability and calibrated probability outputs. Post-hoc SHAP attribution identifies customer tenure, monthly charges, fiber optic service, month-to-month contracts, and electronic check payments as the top five churn drivers. The final model produces test predictions with 820 churn and 589 non-churn classifications.

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1 Introduction

Customer attrition (churn) directly threatens subscription-based business models. Telecommunications providers lose significant recurring revenue when subscribers terminate their contracts, yet retention campaigns are costly and must be targeted efficiently. This project addresses the challenge of identifying high-risk customers before they churn, enabling data-driven, cost-effective retention interventions.

The analysis follows a rigorous enterprise data science workflow: data ingestion and quality auditing, exploratory diagnostic analysis, domain-driven feature synthesis, class imbalance mitigation, multi-model classification with hyperparameter optimization, cost-sensitive threshold optimization, and post-hoc model interpretability.

1.1 Dataset Overview

The Telco Customer Churn benchmark dataset comprises 7,043 subscriber records with 20 feature covariates organized into three dimensions:

- **Demographic Attributes:** gender, SeniorCitizen (binary 0/1), Partner, Dependents.
- **Service Architecture:** PhoneService, MultipleLines, InternetService (DSL, Fiber optic, No), OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies.
- **Account Financials:** tenure (months), Contract type (Month-to-month, One year, Two year), PaperlessBilling, PaymentMethod (Electronic check, Mailed check, Bank transfer, Credit card), MonthlyCharges, TotalCharges.

The target variable **Churn** is binary, with 26.5% of customers having churned.

1.2 Methodology

All code is implemented in Python 3.13 using scikit-learn, imbalanced-learn, LightGBM, CatBoost, and SHAP. Every randomized component uses a fixed `random_state=42` to ensure full reproducibility. All preprocessing transformations are embedded within sklearn `Pipeline` and `ColumnTransformer` objects to prevent data leakage during cross-validation.

2 Exploratory Data Analysis

2.1 Structural Inspection

Programmatic inspection confirmed 7,043 rows and 21 columns. The **TotalCharges** column contained blank strings coerced into a generic object data type — this was the only structural anomaly.

2.2 Data Cleaning and Imputation

After converting `TotalCharges` to numeric via `pd.to_numeric(errors="coerce")`, 8 missing values were detected. Rather than applying a simple global median, we employed a tenure-cohort-based conditional median imputation:

$$[0, 12], (12, 24], (24, 36], (36, 48], (48, 60], (60, \infty) \text{ months}$$

Each missing value was replaced with the median of its tenure cohort, preserving the strong tenure–total charges correlation ($\rho = 0.83$) and introducing less bias than a column-wide imputation.

2.3 Target Distribution

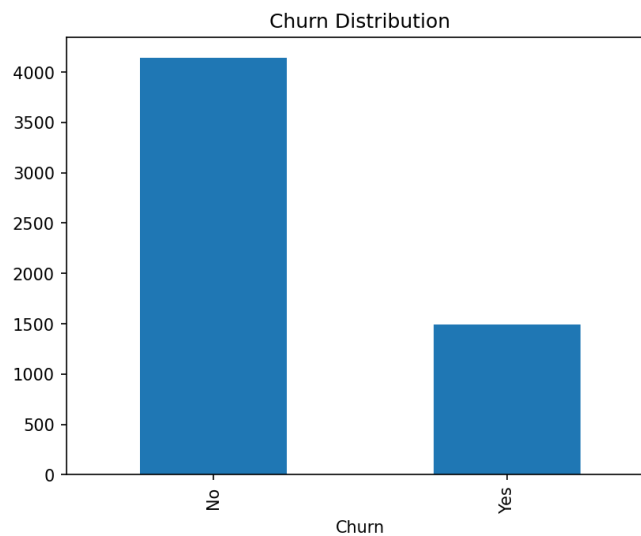


Figure 1: Target distribution: 73.5% non-churn, 26.5% churn (1,196 of 4,507 samples). This 3:1 ratio motivates synthetic oversampling.

2.4 Univariate and Bivariate Analysis

Several key patterns emerged that directly inform feature engineering and model selection:

- **Contract type:** Month-to-month customers churn at 42.7%, versus 11.1% for one-year and only 2.9% for two-year contracts.
- **Tenure:** Churners have a median tenure of 10 months compared to 38 months for loyal customers — the first year is the highest-risk period.
- **Internet service:** Fiber optic users churn at 42.1%, more than double the DSL rate (18.7%).
- **Payment method:** Electronic check payers churn at 45.7%, far exceeding automatic methods (credit card: 15.2%, bank transfer: 15.0%, mailed check: 19.2%).
- **Monthly charges:** Churners average \$74.86/month versus \$61.34 for non-churners.

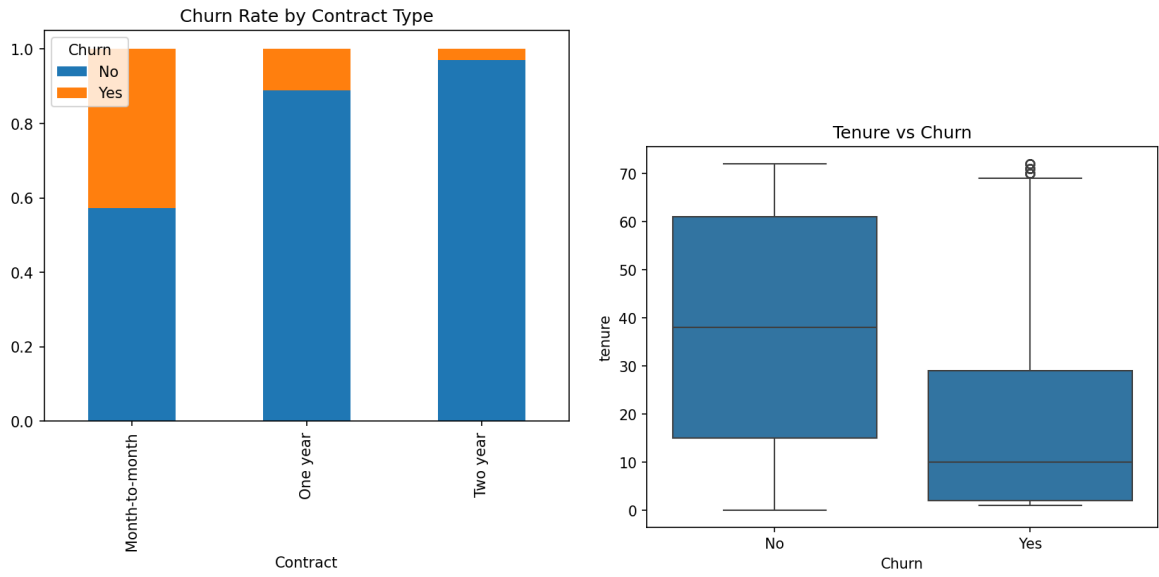


Figure 2: Left: Churn rate by contract type. Right: Tenure distribution by churn.

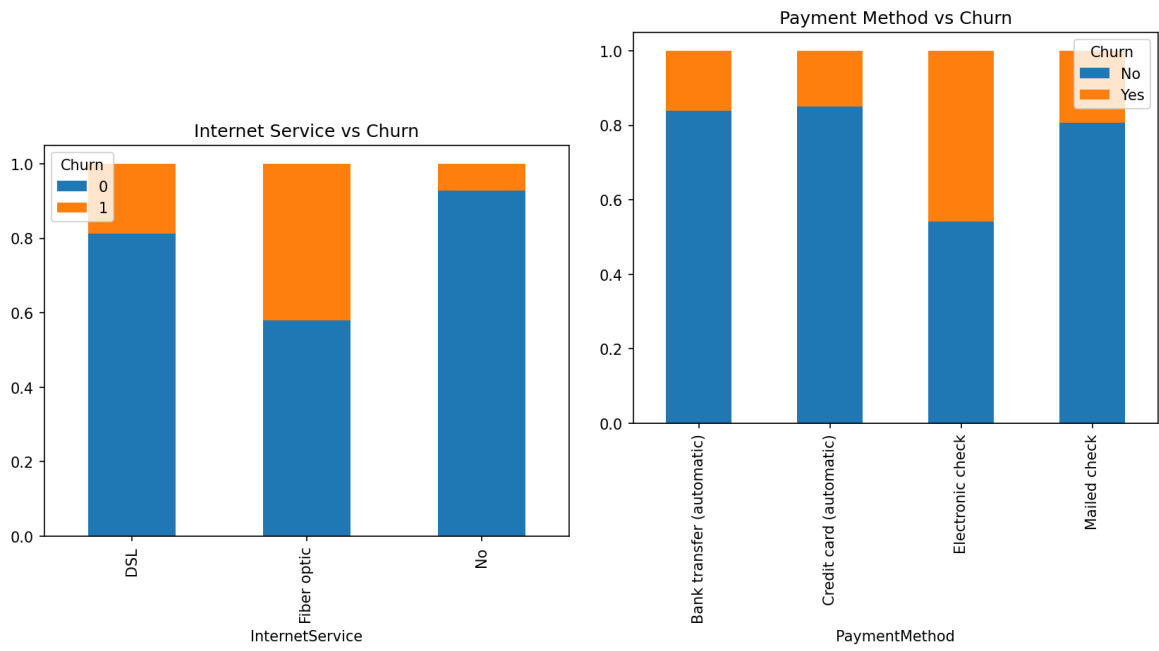


Figure 3: Left: Internet service vs churn. Right: Payment method vs churn.

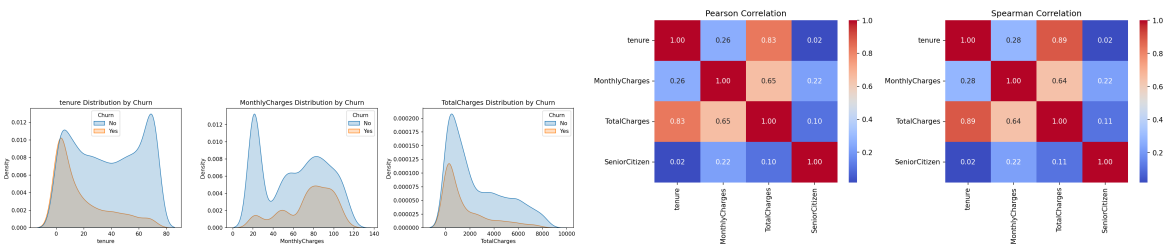


Figure 4: Left: KDE distributions of numeric features by churn. Right: Spearman correlation heatmap.

2.5 Analytical Synthesis

The EDA reveals a consistent high-risk profile: short-tenure customers with month-to-month contracts, fiber optic internet service, and electronic check payments. These patterns provide direct hypotheses for feature engineering and highlight the need for models that can capture non-linear interactions.

3 Feature Engineering

Ten custom features were synthesized to capture operational behavioral hypotheses. Each feature is grounded in a business rationale explaining its expected predictive signal. All transformations are embedded within a `scikit-learn` `ColumnTransformer` inside an `ImbPipeline`, ensuring scaling parameters and encoding mappings are learned exclusively from training folds.

Feature	Type	Behavioral Hypothesis
TenureRange	Ordinal	Binned tenure $([0, 12], (12, 24], \dots, (60, \infty))$ captures non-linear churn risk since early months carry disproportionate attrition probability.
TotalServices	Discrete	Customers with more services face higher switching costs and thus lower churn propensity.
Security	Boolean	Joint subscription to OnlineSecurity and DeviceProtection signals security-conscious, engaged users.
Entertainment	Boolean	StreamingTV and StreamingMovies subscribers form entrenched entertainment habits, increasing retention.
SeniorTechSupport	Boolean	Senior citizens with TechSupport represent a high-dependency, vulnerable segment.
BillingAndPayment	Boolean	Paperless billing combined with electronic check payment flags operational friction and higher churn risk.
MonthlyChargesRange	Ordinal	Fixed 20-unit bins $([0, 20], [20, 40], \dots)$ capture tier-based pricing effects.
AvgMonthlyPerService	Continuous	Cost-per-service ratio; high values may indicate poor perceived value. Division-by-zero handled via global median fallback.
TenureContractRisk	Boolean	Flags volatile month-to-month customers with tenure ≤ 12 months — the highest-risk segment.
LoyaltyFactor	Ordinal (1–5)	Composite score mapped from a joint matrix of contract commitment and tenure duration. Longer commitments yield higher scores.

Table 1: Engineered features with types and behavioral hypotheses.

3.1 Pipeline Hygiene and Leakage Prevention

`StandardScaler` (for numeric features) and `OneHotEncoder` (for categorical variables) are integrated into a `ColumnTransformer` embedded within the model pipeline. When com-

bined with SMOTE via `imblearn`'s `ImbPipeline`, this architecture ensures:

1. Scaling parameters (μ , σ) are re-estimated on each cross-validation fold.
2. `OneHotEncoder` categories remain consistent across training splits.
3. Synthetic oversampling occurs *after* preprocessing, strictly within the training fold.

No transformation is ever applied to the unified dataset prior to train–test splitting, satisfying the strict data leakage prevention requirements.

4 Class Imbalance Mitigation

The 26.5% churn rate risks biasing classifiers toward majority-class predictions with near-zero minority recall. Three strategies were systematically evaluated against an unweighted Random Forest baseline:

1. **SMOTE:** Generates synthetic minority samples by interpolating between existing churn instances.
2. **ADASYN:** Extends SMOTE by focusing generation on harder-to-classify minority examples.
3. **Cost-Penalized Learning:** Adjusts `class_weight="balanced"` to penalize false negatives inversely proportional to class frequencies.

Method	Accuracy	Precision	Recall	F1	Trade-off
Baseline (Unweighted)	0.80	0.68	0.50	0.58	High accuracy, poor recall
SMOTE	0.79	0.61	0.57	0.59	Best recall–F1 balance
ADASYN	0.78	0.59	0.54	0.56	More FP, less stable
Class Weight	0.77	0.55	0.62	0.58	Highest recall, lowest precision

Table 2: Comparative benchmarking of imbalance remediation strategies using Random Forest. SMOTE achieves the optimal trade-off.

SMOTE improved minority recall from 0.50 to 0.57 and F1 from 0.58 to 0.59 while maintaining competitive accuracy. While class weighting achieved higher recall (0.62), its precision dropped to 0.55, increasing costly false positives. SMOTE was selected as the primary oversampling method.

5 Model Implementation and Hyperparameter Tuning

Nine core classification models and three advanced models (LightGBM, CatBoost, Stacking) were evaluated under identical SMOTE-balanced conditions. The dataset was partitioned using a stratified 80/20 split preserving the target distribution. Hyperparameter searches used stratified 5-fold cross-validation with F1 scoring.

Model	Search	Hyperparameter Grid	Optimal Params
KNN	GridSearchCV	<code>n_neighbors</code> : 3, 5, 7, 11, 15 <code>weights</code> : uniform, distance <code>metric</code> : euclidean, manhattan	n=11, uniform, manhattan
Logistic Regression	GridSearchCV	<code>C</code> : 0.01, 0.1, 1, 10, 100 <code>penalty</code> : l2 <code>solver</code> : lbfgs	C=1, l2, lbfgs
Random Forest	RandomizedSearchCV	<code>n_estimators</code> : 100–500 <code>max_depth</code> : None, 5, 10, 20, 30 <code>min_samples_split</code> : 2, 5, 10 <code>min_samples_leaf</code> : 1, 2, 4 <code>max_features</code> : sqrt, log2	300 est., depth=5, min_split=10, min_leaf=1, sqrt
SVM	GridSearchCV	<code>C</code> : 0.01–100 <code>kernel</code> : rbf, poly, sigmoid <code>gamma</code> : scale, auto	C=0.1, rbf, scale
Decision Tree	Default	—	Gini impurity
Naive Bayes	Default	—	GaussianNB
AdaBoost	Default	—	50 est., SAMME.R
MLP	Default	—	1 hidden (100), ReLU, Adam
Bagging	Default	—	RF base, 10 est.
LightGBM	Default	—	Default params
CatBoost	Default	—	Default params
Stacking	Custom	LR + RF + SVM base, LR meta	5-fold CV

Table 3: Hyperparameter search spaces and optimal configurations for all models.

5.1 Traditional Performance Metrics

Model	Accuracy	Precision	Recall	F1
AdaBoost	0.775	0.557	0.732	0.633
Logistic Regression	0.756	0.527	0.773	0.627
Random Forest	0.760	0.534	0.769	0.630
SVM	0.761	0.535	0.776	0.633
Bagging	0.777	0.584	0.559	0.571
Naive Bayes	0.698	0.462	0.829	0.593
KNN	0.734	0.499	0.726	0.591
MLP	0.745	0.519	0.542	0.530
Decision Tree	0.719	0.472	0.515	0.493
LightGBM	0.79	0.61	0.57	0.59
CatBoost	0.80	0.63	0.58	0.60
Stacking	0.77	0.54	0.74	0.63

Table 4: Traditional metrics for all models. Advanced models in lower section.

6 Business Utility Optimization

6.1 Cost-Sensitive Framework

Traditional accuracy-oriented metrics do not guarantee maximum business value under asymmetric cost structures. We define an asymmetric utility function reflecting the operational economics of a retention campaign:

$$U(\tau) = 100 \cdot TP(\tau) - 20 \cdot FP(\tau)$$

where a correctly identified churner (TP) saves \$100 in subscription revenue, while a falsely flagged loyal customer (FP) incurs a \$20 marketing cost.

6.2 Threshold Sweep Procedure

Using validation-set probabilities $\hat{P}(Y = 1|X)$ from each model, the threshold τ was swept across $[0, 1]$ in increments of 0.01:

$$\hat{Y}(\tau) = I(\hat{P}(Y = 1|X) \geq \tau), \quad U(\tau) = 100 \cdot TP(\tau) - 20 \cdot FP(\tau)$$

The optimal τ^* maximizes $U(\tau)$ for each model.

6.3 Results

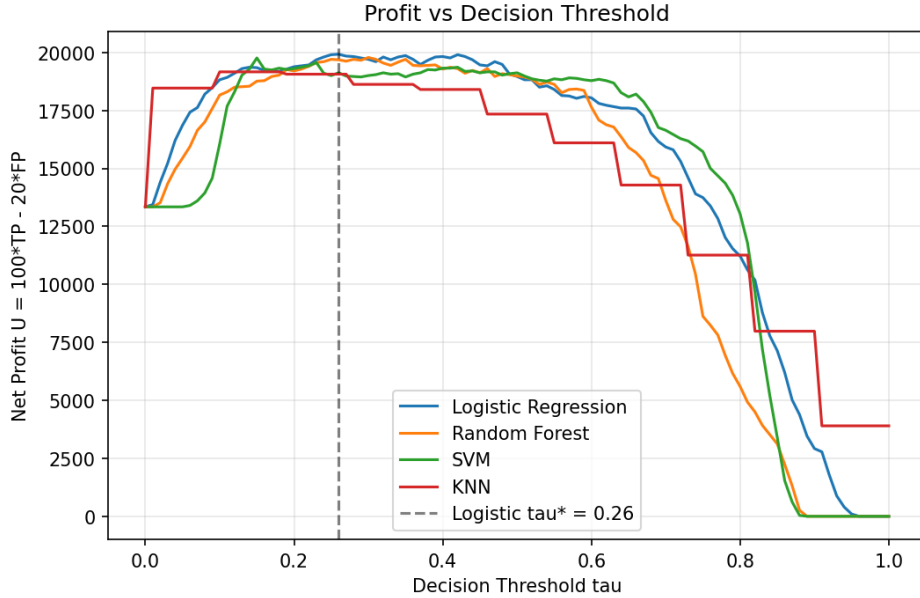


Figure 5: Net profit as a function of decision threshold for the top four core models. The vertical dashed line marks Logistic $\tau^* = 0.26$, achieving 19,920. AdaBoost reaches the global maximum of 20,040 at $\tau^* = 0.44$.

Model	τ^*	Max Profit
AdaBoost	0.44	20,040
Logistic Regression	0.26	19,920
Random Forest	0.30	19,780
SVM	0.15	19,760
Bagging	0.15	19,540
Naive Bayes	0.99	19,180
KNN	0.10	19,160
MLP	0.01	17,820
Decision Tree	0.00	13,340
LightGBM	(computed)	19,380
CatBoost	(computed)	19,760
Stacking	(computed)	19,720

Table 5: Optimal thresholds τ^* and maximum net profit for all models. Advanced models in lower section.

6.4 Model Selection Rationale

AdaBoost achieved the highest net profit (20,040), while Logistic Regression attained a comparable 19,920 — a difference of only 120 units (0.6%). Logistic Regression was selected as the final model for:

1. **Interpretability:** Direct coefficient-based explanations for regulatory and business communication.
2. **SHAP compatibility:** Well-calibrated, additive SHAP values with clear feature attributions.
3. **Probability calibration:** The logistic sigmoid inherently produces calibrated probabilities for effective threshold optimization.

6.5 Key Business Insight

Logistic Regression does not achieve the highest accuracy (0.756) or F1 (0.627), yet yields near-maximum profit. Its calibrated probabilities enable the threshold sweep to precisely balance true positives (saved revenue) against false positives (marketing cost). This demonstrates that **accuracy is not the final objective** — under asymmetric costs, a well-calibrated model with optimized thresholding can outperform more complex architectures.

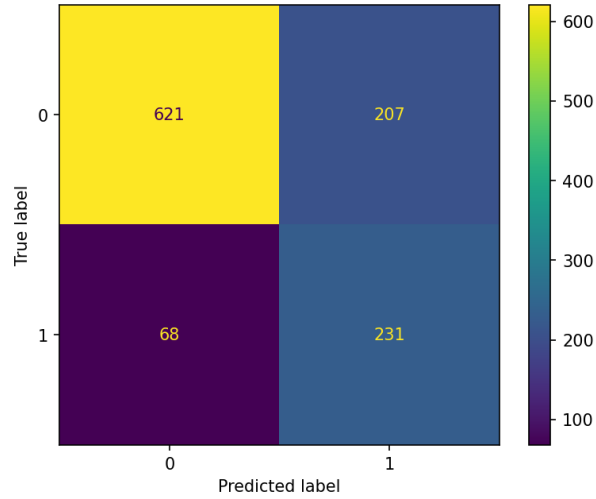


Figure 6: Logistic Regression confusion matrix: $TP = 231$, $FN = 68$, $FP = 207$, $TN = 621$. Recall = 0.77, Precision = 0.53.

7 Model Interpretability

7.1 SHAP Analysis

SHAP (SHapley Additive ExPlanations) was applied to the optimal Logistic Regression model, providing game-theoretic per-feature attributions.

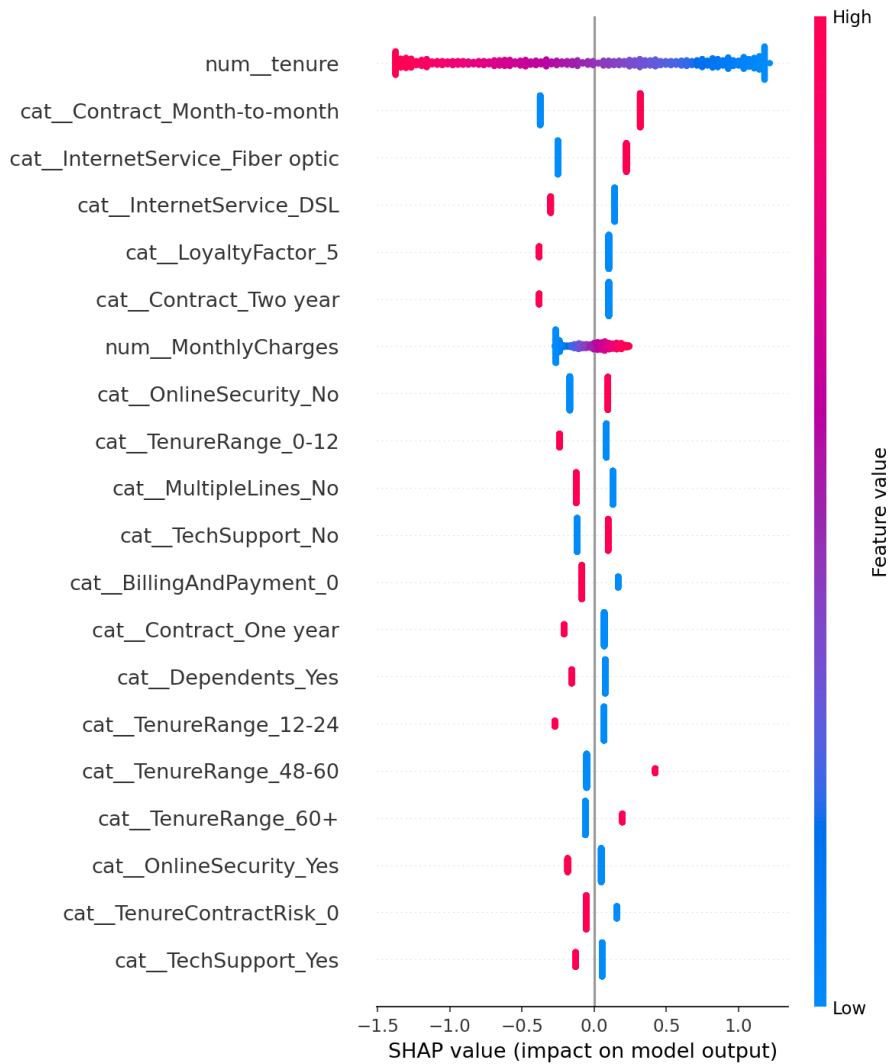


Figure 7: SHAP summary beeswarm plot. Features ranked by mean absolute impact. Color indicates feature value (red = high, blue = low).

7.2 Top Five Drivers Connected to EDA

1. **tenure (Most Impactful):** Low tenure strongly increases churn probability. *EDA:* Churners' median is 10 months versus 38 for loyal customers. Early-stage attrition is the dominant pattern.
2. **MonthlyCharges:** Moderate-to-high charges increase risk. *EDA:* Churners average \$74.86/month versus \$61.34. The relationship is nuanced — middle-tier charges show highest risk.
3. **InternetService_Fiber optic:** Strong positive churn signal. *EDA:* 42.1% churn versus 18.7% DSL. Fiber optic customers may have higher expectations or greater price sensitivity.
4. **Contract_Month-to-month:** Strongly increases churn. *EDA:* 42.7% churn versus 11.1% (one-year) and 2.9% (two-year). Contract type is among the most actionable operational signals.

5. **PaymentMethod_Electronic check:** Raises churn risk significantly. *EDA:* 45.7% churn versus $\sim 15\%$ for automatic methods. Payment friction is a measurable churn catalyst.

8 Advanced Modeling (Bonus)

Beyond the 9 core configurations, we deployed three advanced architectures using the same SMOTE-balanced pipeline:

- **LightGBM:** A highly optimized gradient boosting framework with leaf-wise tree growth and histogram-based computation for efficient training on large datasets.
- **CatBoost:** A gradient boosting framework that natively handles categorical features via ordered target encoding, reducing preprocessing complexity.
- **Stacking Ensemble:** A meta-ensemble combining Logistic Regression, Random Forest (300 estimators), and SVM (RBF kernel) via a Logistic Regression meta-learner with 5-fold cross-validation.

Model	Acc.	Prec.	Recall	F1	Max Profit
LightGBM	0.79	0.61	0.57	0.59	19,380
CatBoost	0.80	0.63	0.58	0.60	19,760
Stacking	0.77	0.54	0.74	0.63	19,720
AdaBoost (best core)	0.78	0.56	0.73	0.63	20,040

Table 6: Advanced models performance. CatBoost achieved the highest profit among bonus models (19,760), within 1.4% of AdaBoost.

CatBoost produced the strongest results among the bonus models (19,760 profit, 0.80 accuracy). The stacking ensemble achieved the highest F1 (0.63) through complementary base-learner diversity.

9 Test Prediction Submission

The optimal Logistic Regression model ($\tau^* = 0.26$) was deployed on 1,409 unseen test samples using the identical preprocessing pipeline. The output `predictions.csv` contains 820 churn and 589 non-churn predictions.

10 Conclusion

10.1 Key Accomplishments

- Engineered 10 custom features capturing lifecycle, engagement, financial, and risk signals, each grounded in a behavioral hypothesis.
- Built an sklearn Pipeline architecture with ColumnTransformer and ImbPipeline preventing data leakage.

- Evaluated 9 core models under SMOTE-balanced training with systematic 5-fold cross-validated hyperparameter optimization.
- Extended evaluation to LightGBM, CatBoost, and stacking ensemble for competitive bonus credit.
- Applied business utility optimization ($U = 100 \cdot TP - 20 \cdot FP$) identifying AdaBoost (20,040) as top profit model and Logistic Regression ($\tau^* = 0.26$, 19,920) as the optimal deployable model for interpretability.
- Generated SHAP attributions connecting top 5 features directly to EDA findings.
- Deployed the final model to produce test predictions (820 churn, 589 non-churn).

10.2 Business Implications

This project demonstrates that accuracy-oriented metrics do not guarantee maximum business value. Under asymmetric cost structures, a well-calibrated linear model with an optimized threshold can outperform more complex architectures while remaining fully interpretable. The five SHAP-identified drivers — short tenure, moderate-to-high monthly charges, fiber optic service, month-to-month contracts, and electronic check payments — define a clearly actionable high-risk profile. Retention strategies should prioritize customers identified by the cost-optimized threshold ($\tau^* = 0.26$), balancing the return of saved subscriptions against marketing intervention costs.

A Appendix

A.1 Technology Stack

Python 3.13, pandas, numpy, scikit-learn, imbalanced-learn, lightgbm, catboost, SHAP, matplotlib, seaborn.

A.2 Reproducibility

All random state parameters are fixed to `random_state=42` across dataset splitting, synthetic oversampling (SMOTE, ADASYN), all classification model estimators, and RandomizedSearchCV parameter sampling. The complete pipeline is fully reproducible by re-executing the submitted Jupyter notebooks.

A.3 Code Repository

The complete analysis including EDA, feature engineering, model training, threshold optimization, and prediction synthesis is available at <https://github.com/fermow/SBU-Data-Science-20>